



38. (Python Question) Global scope

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What will be the output of `print(foo_1())` and `print(foo_2())` respectively?



```
def foo_1():  
    return total + 1
```

```
def foo_2():  
    total += 1  
    return total
```

```
total = 0
```

Pick **ONE** option

☐ 1, Error

☐ 0, 1

☐ 1, 1

☐ None of the above



ALL



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19. (C Question) Pointers II

Predict the output of following C program:

```
# include <stdio.h>

void main ( )

{
    char *s[ ] = {"black", "white", "yellow", "violet"};
    char **ptr[] = {s+3, s+2, s+1, s}, ***p;

    p = ptr;

    ***p;

    printf("%s\n", *--***p + 3);
}
```

Pick **ONE** option

☐ ow☐ black

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36. (Coding) Tom & Jerry in a Maze

After decades of chasing Jerry, Tom wants to make peace. Being skeptical, Jerry hides in an $n \times n$ maze. Tom decides the best way to make Jerry take him seriously is to collect all of the cheese pieces in the maze and give them to Jerry as a gift. A *move* is considered to be a single step from the current position to some adjacent position in the maze. Tom can move in all four directions within the maze: up ($\uparrow$), down ($\downarrow$), left ($\leftarrow$), and right ($\rightarrow$).

Complete the *minMoves* function in your editor. It has 3 parameters:

1. A 2D array of integers, *maze*, denoting the maze where Jerry is hiding.
2. An integer, *x*, denoting the x-coordinate for Jerry's location.
3. An integer, *y*, denoting the y-coordinate Jerry's location.

Each cell in *maze* is labeled as follows:

- A path cell is represented by a 0.
- A blocked cell (wall) is represented by a 1.
- A cheese cell is represented by a 2. Tom can move through a cheese cell just as he would a regular path cell.

Language C++

```
1 > #include
10 |
11 // Comple
12 int minMo
13
14
15 }
16 > int main()
```

Test Results



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Explanation

We perform the following $n = 5$ calculations:

0. $722/148 + 360/176$ evaluates to the reduced fraction $2818/407$.

1. $978/1212 + 183/183$ evaluates to the reduced fraction $365/202$.

2. $358/472 + 301/417$ evaluates to the reduced fraction $145679/98412$.

3. $780/309 + 684/988$ evaluates to the reduced fraction $4307/1339$.

4. $258/840 + 854/686$ evaluates to the reduced fraction $1521/980$.

Thus, we return the array `["2818/407", "365/202", "145679/98412", "4307/1339", "1521/980"]` as our answer.

Sample Input 1

```
10
241/26+612/398
278/348+246/157
156/53+952/760
263/418+560/304
540/28+44/295
636/78+354/868
68/415+1041/807
147/269+224/844
759/120+370/72
654/162+476/307
```

Sample Output 1

Language C++

```
10 vector<string>
11 for
12
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40 }
```

Test Results



ALL



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9. (Prog. Concept Question) Guess the Data Structure

In what kind of storage structure for strings can one easily insert, delete, concatenate and rearrange substrings?

Pick **ONE** option

☐ fixed length storage structure

☐ variable length storage with fixed maximum

☐ linked list storage

☐ array type storage



Type here to search



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Sample Output 1

55915/5174
64627/27318
21127/5035
1033/418
40133/2065
48305/5642
162297/111635
46081/56759
4127/360
46315/8289

Explanation

We perform the following $n = 10$ calculations:

0. $241/26 + 612/398$ evaluates to the reduced fraction $55915/5174$.
1. $278/348 + 246/157$ evaluates to the reduced fraction $64627/27318$.
2. $156/53 + 952/760$ evaluates to the reduced fraction $21127/5035$.
3. $263/418 + 560/304$ evaluates to the reduced fraction $1033/418$.
4. $540/28 + 44/295$ evaluates to the reduced fraction $40133/2065$.
5. $636/78 + 354/868$ evaluates to the reduced fraction $48305/5642$.
6. $68/415 + 1041/807$ evaluates to the reduced fraction $162297/111635$.

Language C++

Test Results

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ALL

**Output Format**

Your function must return an integer denoting the minimum number of moves Tom must make to collect all of the maze's cheese and deliver it to Jerry; if the task is not possible, return -1. This is printed to stdout by the locked stub code in your editor.

Sample Input 0

The following arguments are passed to your function:

```
maze = {{0, 2, 0}, {0, 0, 1}, {1, 1, 1}}
```

```
x = 1
```

```
y = 1
```

Sample Output 0

```
2
```

Sample Input 1

The following arguments are passed to your function:

```
maze = {{0, 1, 0}, {1, 0, 1}, {0, 2, 2}}
```

```
x = 1
```

```
y = 1
```

Sample Output 1

```
-1
```

Sample Input 2

Test Results

Language C++

```
1 > #include
10
11 // Comp
12 int minM
13
14
15 }
16 > int main()
```

0m left

Language C++

Explanation

We perform the following $n = 10$ calculations:

0. $241/26 + 612/398$ evaluates to the reduced fraction $55915/5174$.
1. $278/348 + 246/157$ evaluates to the reduced fraction $64627/27318$.
2. $156/53 + 952/760$ evaluates to the reduced fraction $21127/5035$.
3. $263/418 + 560/304$ evaluates to the reduced fraction $1033/418$.
4. $540/28 + 44/295$ evaluates to the reduced fraction $40133/2065$.
5. $636/78 + 354/868$ evaluates to the reduced fraction $48305/5642$.
6. $68/415 + 1041/807$ evaluates to the reduced fraction $162297/111635$.
7. $147/269 + 224/844$ evaluates to the reduced fraction $46081/56759$.
8. $759/120 + 370/72$ evaluates to the reduced fraction $4127/360$.
9. $654/162 + 476/307$ evaluates to the reduced fraction $46315/8289$.

Thus, we return the array `["55915/5174", "64627/27318", "21127/5035", "1033/418", "40133/2065", "48305/5642", "162297/111635", "46081/56759", "4127/360", "46315/8289"]` as our answer.

Test Results

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Tom's initial position is $(0, 0)$. Your function must return an integer denoting the minimum number of moves that it will take for him to collect all the cheese and deliver it to Jerry at (x, y) . If the task is not possible, return -1 .

Input Format

The locked stub code in your editor reads the following input from stdin and passes it to your function:

The first line contains an integer, n , denoting the number of rows in *maze*.

The second line contains an integer, n , denoting the number of columns in *maze*.

Each line i of the n subsequent lines (where $0 \leq i < n$) contains n space-separated integers describing the respective elements of row i in *maze*.

The next line contains an integer, x , denoting the x -coordinate where Jerry is located in *maze*.

The next line contains an integer, y , denoting the y -coordinate where Jerry is located in *maze*.

Constraints

- $1 \leq n \leq 100$
- $0 \leq \text{the number of cheese pieces} \leq 10$
- $1 \leq x, y \leq n$

Output Format

Your function must return an integer denoting the minimum number of moves Tom must make to collect all of the maze's cheese and deliver it to Jerry; if the task is not possible, return -1 . This is

Language C++

```
1 > #include
10
11 // Comp
12 int min
13
14
15 }
16 > int main
```

Test Results



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21. (C++ Question) Adding two objects in C++

If we have three instances of a class "SignedInt", with two properties:

1. Value : Integral value
2. Sign : positive/negative

No other methods have been defined for this class other than the GET and SET methods.
Is the statement "C = A + B;" valid/legal? If yes, state reasons. If not, then rectify the issue.

Pick **ONE** option

☐

Yes, It is valid because the compiler calls the getter and setter functions.

☐

Yes, because the + operator calls the copy constructor of both A and B.

☐

No, we need to overload the copy constructor of both A and B.

☐

No, we need to overload the + operator of both A and B.

Clear Selection



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Sample Output 1

-1



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Sample Input 2

The following arguments are passed to your function:

```
maze = {{0, 2, 0}, {1, 1, 2}, {1, 0, 0}}
```

```
x = 2
```

```
y = 1
```

Sample Output 2

5

Explanation*Sample Case 0:*

The shortest path Tom can take to pick up all the cheese and deliver it to Jerry is $(0, 0) \rightarrow (0, 1) \rightarrow (1, 1)$. Because this involves 2 moves, we return 2.

Sample Case 1:

It is not possible for Tom to reach Jerry, so we return -1.

Sample Case 2:

The shortest path Tom can take to pick up all the cheese and deliver it to Jerry is $(0, 0) \rightarrow (0, 1) \rightarrow (0, 2) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (2, 1)$. Because this involves 5 moves, we return 5.

Test Results

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ALL

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```
char *ptr[] = {s+3, s+2, s+1, s}, *pp;
```

```
p = ptr;
```

```
*++p;
```

```
printf("%s\n", s---*++p + 3);
```

```
}
```

Pick **ONE** option

☐ ow☐ black☐ ck☐ Run time error☐ Compile time error



23. (C++ Question) Exceptions in C++

ALL



Consider the following code snippet below. Will the exception will be caught properly in the try/catch block. ?

```
19  #include <iostream>
    class myException
    {
    public:
        myException(std::string aMsg)
        {
            exMsg = aMsg;
        }
        std::string what()
        {
            return exMsg;
        }
    private:
        std::string exMsg;
23  };
    int main()
    {
24      try
        {
25          throw myException("Default message.");
        }
        catch (std::exception e)
        {
26            std::cout << e.what() << std::endl;
        }
27    }
```



ALL



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What is the output of the following code snippet?

```
class A{
    public:
        A(){
            cout<<"A"<<endl;
        }
};
class B: public A{
    public:
        B(){
            cout<<"B"<<endl;
        }
};
int main(){
    B b;
}
```

Pick **ONE** option

☐ B☐ A B☐ B A☐ None of the above



ALL



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25. (C++ Question) Default methods in a class in C++

Which of these methods are present by default in a class? (Select all that apply)

Pick **ONE OR MORE** options

☐

Constructor

☐

Destructor

☐

Assignment operator

☐

Addition operator

☐

Copy Constructor

[Clear Selection](#)

26. (C++ Question) Virtual functions

ALL

24. (C++ Question) Copy constructor

①

What would be the output of the following code snippet?

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```
#include <iostream>
```

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```
class A{  
public:
```

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```
    int a;
```

```
    int* b;
```

```
    A(int x, int* y) : a(x), b(y){}
```

```
    A(const A& Ain){
```

```
        this->a = Ain.a;
```

23

```
    }
```

```
};
```

24

```
int main(){
```

```
    A p(10, NULL);
```

```
    A q(p);
```

25

```
    if(q.b)
```

```
        std::cout<<"true";
```

```
    else
```

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```
        std::cout<<"false";
```

```
}
```

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Pick **ONE** option

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22. (C++ Question) Function in a Structure in C ++

It is illegal to define a function within a structure in C++. State True or False

Pick **ONE** option

☐

TRUE

☐

FALSE

☐

Depends on Compiler

Clear Selection

23. (C++ Question) Exceptions in C++

Consider the following code snippet below. Will the exception will be caught properly in the try/catch block. ?

```
#include <iostream>
```



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```
else  
    std::cout<<"false";  
}
```

Pick **ONE** option

☐ true

☐ false

☐ Compiler Error

☐ No Output

Clear Selection

25. (C++ Question) Default methods in a class in C++

Which of these methods are present by default in a class? (Select all that apply)

Pick **ONE OR MORE** options



ALL



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```
b = &d;  
b->show();  
}
```

Pick **ONE OR MORE** options

☐ Prints "Show Function In Derived class called"

☐ Error on line "b = &d;"

☐ Error on line "Base obj;"

☐ Error on line "Derived d;"

[Clear Selection](#)

27. (C++ Question) Templated node class

Select example(s) of templated node class that can be used in a linked list.

Pick **ONE** option



ALL



```
catch (std::exception e)
{
    std::cout << e.what() << std::endl;
}
}
```

Pick **ONE** option

19

☐ Yes, because exception is a more general class.

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☐ Yes, because myException is also implemented in the same scope.

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☐ No, because myException is not a subclass of exception.

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☐ No, because the definition of myException is invalid.

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Clear Selection

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24. (C++ Question) Copy constructor

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What would be the output of the following code snippet?



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27. (C++ Question) Templated node class

Select example(s) of templated node class that can be used in a linked list.

Pick **ONE** option

☐ 1. `template <class T> class node { node* next; T value; }`

☐ 2. `template <class T> class node { public: node* next; T value; }`

☐ 3. `class <template T> class node { node* next; T value; }`

☐ 4. `class <template T> class node { public: node* next; T value; }`

Clear Selection

Continue

8. (Prog. Concept Question) Static Polymorphism

ALL

Which of the following is not a static polymorphism?

Pick **ONE** option

☐ Function overloading

☒ Operator Overloading

☐ Function Overriding

☐ None



ALL



10. (Prog. Concept Question) B-Tree

B - Trees are used commonly for:

Pick **ONE** option

8

☐ efficiently sorting very large files

9

☐ for checking the consistency of relational databases

10

☐ speeding up database search, delete, and insert operations

11

☐ organizing hash tables efficiently

12



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ALL

i

```
struct country India;
```

```
};
```

```
struct world st = { {1, 2, 3}, 'P', 'q', 1.4};
```

```
printf("%d\t%c\t%c\t%f", st.a[1], st.b, st.India.c, st.India.d);
```

```
}
```

Pick **ONE** option

☐ 2 'P' 'q' 1.4

☒ 'P' 'q' 1.4 2

☐ 'q' 1.4 2 'P'

☐ Compiler Error.

11. (Prog. Concept Question) Bitwise Operators

If $n \& (n - 1) == 0$, what can you say about integer n ?

Pick **ONE** option

☐ n is an even number.

☒ n is a power of 2.

☐ n is an odd number.

☐ n is not a power of 2.

Clear Selection



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26. (C++ Question) Virtual functions

ALL

What will be the output of following code?

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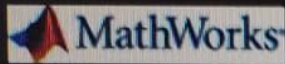
30

```
#include<iostream>
using namespace std;

class Base
{
    public:
        virtual void show() = 0;
        virtual void print() = 0;
};

class Derived:public Base
{
    public:
        void show()
        {   cout << "Show Function in Derived class called"; }
};

int main()
{
    Base obj;
    Base *b;
    Derived d;
    b = &d;
    b->show();
}
```



ALL



42. (Python Question) Inline with recursion

What is the expected output ?

```
List = [3.1416, 1.618, 2.7183, 0.57721]
func = lambda x: (func (x[1:]) + x[:1] if x else [])
print(func (List))
```

Pick **ONE** option

☐ [0.57721, 2.7183, 1.618, 3.1416]

☐ [3.1416, 1.618, 2.7183, 0.57721]

☐ Error, lambda functions can't be called recursively

☐ []

Next Selection



ALL



43. (Python Question) Let's play with variables

What is the expected output?

```
temp = 'Math'
List = []
for i in range(len(temp)):
    List.append(temp)
    temp = 'Physics'
print(List)
```

Pick **ONE** option

☐ ['Math', 'Physics', 'Physics', 'Physics', 'Physics', 'Physics', 'Physics']

☐ ['Math']

☐ Error

☐ None of the above



12. (C Question) Structures in C

ALL

What will be output of following c code?

```
void main()
```

```
{
```

```
struct country
```

```
{
```

```
char c;
```

```
float d;
```

```
};
```

```
struct world
```

```
{
```

```
int a[3];
```

```
char b;
```

```
struct country India;
```

```
};
```

```
struct world st = {{1, 2, 3}, 'P', 'd', 1.4};
```

```
printf("%d\t%c\t%c\t%f", st.a[1], st.b, st.India.c, st.India.d);
```

```
}
```

Pick **ONE** option



ALL



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39. (Python Question) Let us give some numbers part 2

Is the following piece of code valid ?

```
>>> a=2,3,4,5
>>> a
```

Pick **ONE** option

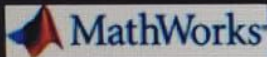
☐ Yes, 2 is printed

☐ Yes, [2,3,4,5] is printed

☐ No, too many values to unpack

☐ Yes, (2,3,4,5) is printed

Clear Selection



ALL



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41. (Python Question) Return from List

What is the expected output?

```
math_constants = [3.1416, 1.618, 2.7183, 0.57721]
Return = math_constants.append(1.30357)
print(math_constants)
print(Return)
```

Pick **ONE** option



[3.1416, 1.618, 2.7183, 0.57721, 1.30357]
TypeError



[3.1416, 1.618, 2.7183, 0.57721, 1.30357]
[3.1416, 1.618, 2.7183, 0.57721, 1.30357]



[3.1416, 1.618, 2.7183, 0.57721, 1.30357]
None



SyntaxError



ALL



40. (Python Question) A map to filtered Lambda

What is the expected output?

```
print(list(map((lambda x:x**3), filter((lambda x:x%2==0), range(5,-5,-2)))))
```

Pick **ONE** option

☐ Error

☐ []

☐ [125, 27, 1, -1, -27]

☐ [125, 64, 27, 8, 1, 0, -1, -8, -27, -64]

Clear Selection